

Recomputed stabilization diagnostics from the validated five-dimensional three-mode stationary density

Analysis-only audit for Section 4.2

15 September 2026

Abstract

We recompute the endpoint-exchange asymmetry, energy-conditioned phase locking, and stationary current balance of the driven three-mode chain from the normalized five-dimensional NESS density trained on projection-free, double-precision trajectories. Every density statistic is restricted to the predeclared Section 4.1 sampled-support mask and compared with an independent held-out trajectory estimate and with the legacy neural density. The new density reproduces the angular marginal and most locking statistics much more closely than the legacy density. Nevertheless, the frozen validation is not a complete pass: the conservative three-times-floor asymmetry gate fails, three of thirty phase-locking comparisons have nonoverlapping confidence intervals, and the masked current balance differs from the trajectory estimate by 2.31 bootstrap standard errors. The global current identity itself passes on the unconditioned held-out trajectories. The resulting Section 4.2 claims must therefore remain support-restricted and diagnostic-specific.

1 Inputs, measures, and frozen gates

The state is $x = (I_1, I_2, I_3, \theta_1, \theta_3)$ with density measured against $dI_1 dI_2 dI_3 d\theta_1 d\theta_3$. We use the fine-step R11 density ensemble with training seeds 8201, 8202, and 8203; the ensemble is the arithmetic mean of three separately normalized densities. The driven case is $(T_1, T_3) = (2, 8)$ and the exact exchange-symmetric control is $(5, 5)$. Uncertainty intervals use 1000 whole-stream bootstrap replicates with seed 20260915042. The independent V6 test set contains 32 driven streams (3,200,032 states) and 16 equilibrium streams (1,600,016 states).

The support mask was fixed from validation trajectories before test evaluation. It keeps the central 0.1–99.9% action ranges, pools I_1 and I_3 when setting their bounds so that the mask is exchange invariant, and retains the full angular torus. It covers 99.420% of the driven held-out probability and 99.403% of the equilibrium held-out probability. The new driven density assigns 99.422% of its mass to the mask; the legacy density assigns 98.412% and has only 3.37% importance-sampling effective sample fraction. No density value is used outside the mask.

For the exchange map

$$\sigma(I_1, I_2, I_3, \theta_1, \theta_3) = (I_3, I_2, I_1, \theta_3, \theta_1),$$

we predeclared the pointwise relative asymmetry

$$A_\sigma(x) = \frac{|\rho(x) - \rho(\sigma x)|}{\rho(x) + \rho(\sigma x)}$$

and the total-variation asymmetry of a fixed 60×60 angular histogram, $TV_\sigma = \frac{1}{2} \sum_{ab} |P_{ab} - P_{ba}|$. The strict asymmetry gate requires all three driven seeds to exceed the equilibrium upper bootstrap floor and the driven lower limit to exceed three times that floor. Phase locking is evaluated in energy terciles fixed from validation data, with 72 periodic angle bins and a fixed half-width $\pi/6$. A density/trajectory locking comparison passes only when the two whole-stream bootstrap intervals overlap.

2 Endpoint-exchange asymmetry

The driven new-density angular TV is 4.00 times the equilibrium point estimate, and its three seed values lie in the narrow range $[0.11405, 0.11567]$. It also agrees with the independent held-out value 0.11587. Thus the observed breaking is reproducible and not a legacy-model effect. Under the deliberately conservative frozen gate, however, the ratio of the driven lower limit to the equilibrium upper floor is only 2.435, not the required 3. The first seed-consistency sub-gate passes, whereas the three-times-floor sub-gate **FAIL**. This result must be reported as a failed formal gate, not converted into a pass by changing the threshold.

Table 1: Exchange-asymmetry diagnostics on the support mask. Brackets give the percentile range of the resampled histogram estimator. Because total variation is nonnegative, this resampling distribution is positively biased at equilibrium and serves as a conservative measured floor.

case	source	mean A_σ	median	p90	maximum	angular TV
driven	new density	0.17961	0.11529	0.44708	0.99894	0.11489 [0.11369,0.11959]
driven	legacy density	0.13620	0.08257	0.35218	0.77445	0.09920 [0.09690,0.10866]
driven	held-out trajectory	—	—	—	—	0.11587 [0.11365,0.12120]
equilibrium	new density	1.30×10^{-16}	0	4.44×10^{-16}	7.11×10^{-15}	0.02871 [0.03414,0.04668]
equilibrium	legacy density	0.00633	0.00510	0.01332	0.04368	0.03079 [0.03578,0.04916]
equilibrium	held-out trajectory	—	—	—	—	0.02747 [0.03276,0.04545]

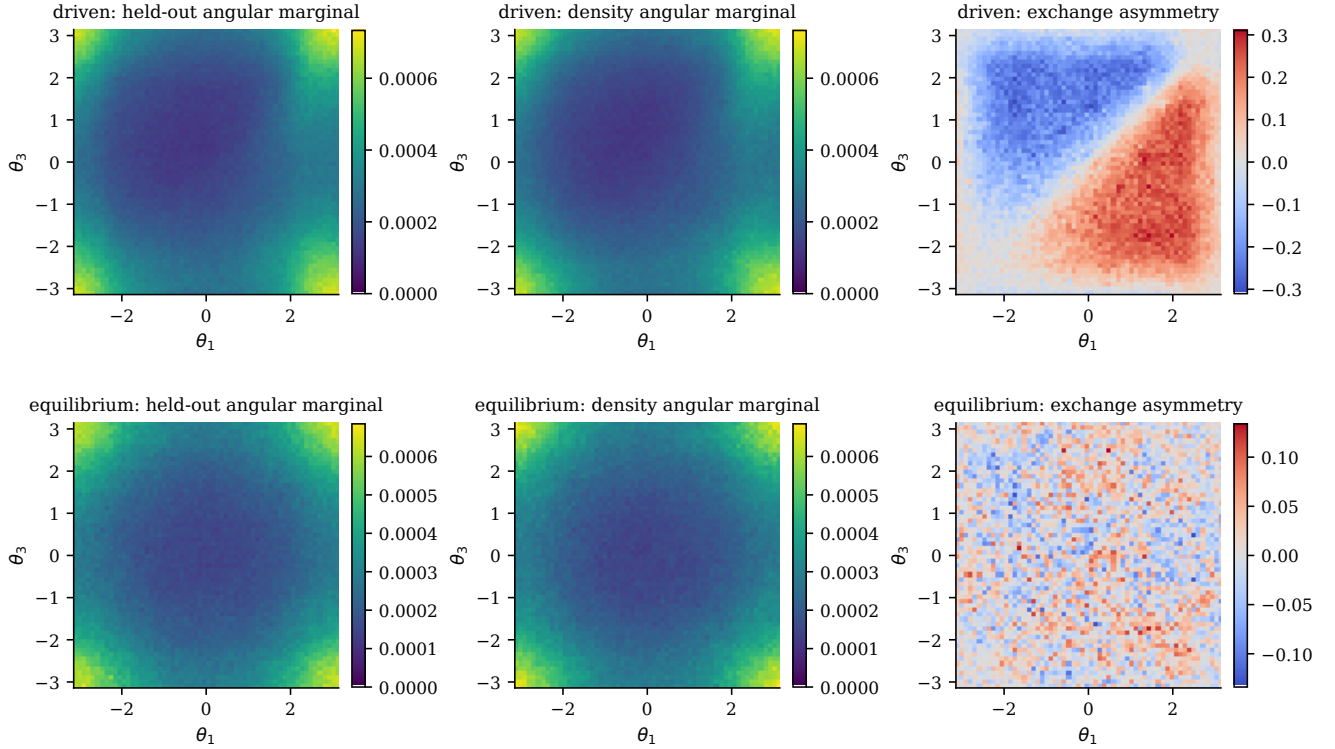


Figure 1: Angular marginals and endpoint-exchange asymmetry. The driven validated density closely tracks the held-out trajectory marginal and exhibits a coherent antisymmetric structure. The equilibrium panel shows the finite histogram floor rather than physical symmetry breaking.

3 Energy-conditioned phase locking

The high-energy branch used without tuning is

$$\theta_0 = \pi - \arcsin \left[\frac{\sqrt{4\gamma^2 + 3} - \gamma}{2(1 + \gamma^2)} \right] = 2.19120 \text{ rad} = 125.547^\circ,$$

with $\gamma = 0.1$. Table 2 gives representative fixed-window probabilities. The complete 360-row table, including circular means, resultants, histogram modes, all three model seeds, both physical cases, and the legacy density, is supplied as `analysis/phase_locking.csv`.

Table 2: Driven conditional locking probabilities. Columns are held-out trajectory / validated density / legacy density. The first row in each sector is $P(|\theta_1 - \theta_0| < \pi/6)$, the second is $P(|\theta_3 + 2\pi/3| < \pi/6)$, and the third requires both angles to lie near θ_0 .

energy	statistic	held-out	new density	legacy density
low	θ_1 near θ_0	0.20413	0.20432	0.20060
low	θ_3 near $-2\pi/3$	0.19329	0.19378	0.19146
low	both near θ_0	0.03906	0.03819	0.03750
middle	θ_1 near θ_0	0.21739	0.21840	0.21150
middle	θ_3 near $-2\pi/3$	0.20279	0.20505	0.20104
middle	both near θ_0	0.03984	0.03928	0.03769
high	θ_1 near θ_0	0.25047	0.24469	0.22301
high	θ_3 near $-2\pi/3$	0.21893	0.21817	0.20742
high	both near θ_0	0.04206	0.04182	0.03842

The rebuilt density agrees with the direct estimate in 27 of 30 predeclared driven comparisons. The three failures are retained in Table 3. Their absolute discrepancies are small but statistically resolved by the large held-out sample. In particular, the high-energy θ_1 locking probability moves from 0.22301 under the legacy density to 0.24469 under the new density, substantially closer to the direct 0.25047, but its intervals still do not overlap.

Table 3: All failed new-density versus held-out phase comparisons.

sector	angle	statistic	density [95%]	trajectory [95%]	difference	z_{approx}
low	θ_3	circular mean	-3.0299 [-3.0549, -3.0077]	-3.0650 [-3.0828, -3.0444]	0.03514	2.26
middle	θ_1	circular mean	2.8894 [2.8698, 2.9084]	2.9194 [2.8997, 2.9360]	-0.02999	-2.21
high	θ_1	locking probability	0.24469 [0.24258, 0.24706]	0.25047 [0.24800, 0.25332]	-0.00578	-3.26

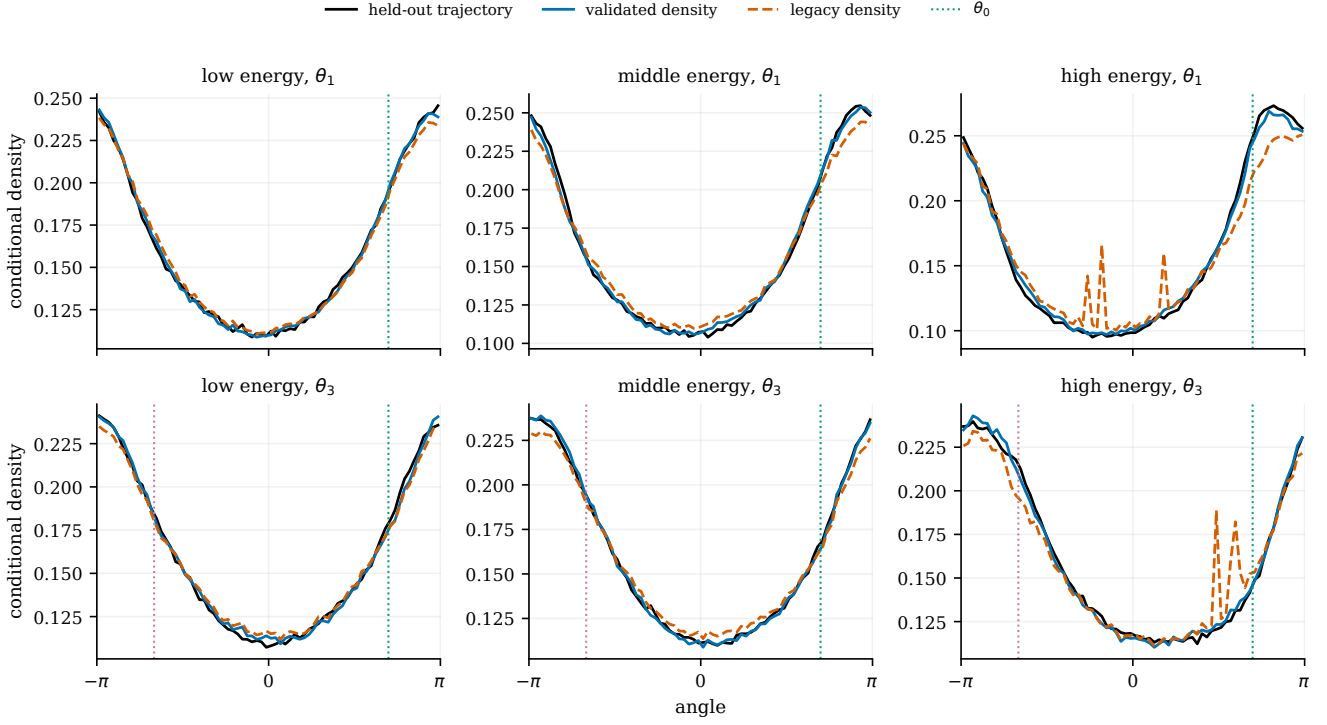


Figure 2: Driven angular distributions conditioned on validation-fixed energy terciles. The validated density follows the held-out curves closely; the legacy density exhibits visible high-energy sampling noise and larger departures. Vertical lines mark the fixed high- and low-energy references.

4 Current balance and scope audit

For the middle action, the state observables are

$$J_{12} = I_1 I_2 \sin \theta_1, \quad J_{23} = I_2 I_3 \sin \theta_3.$$

Stationarity implies the *unconditioned* identity $\langle J_{12} + J_{23} \rangle = 0$. The original frozen protocol mistakenly required this identity after conditioning on the support mask. That conditioned expectation need not vanish because the mask introduces a selection boundary. We preserve the original result and add a post-hoc scope audit rather than silently changing it.

Table 4: Driven current estimates on the common support mask. These values test density reconstruction, not the unconditioned zero identity.

source	$\langle J_{12} \rangle$	$\langle J_{23} \rangle$	sum
held-out trajectory	0.23057 [0.22607,0.23501]	-0.22865 [-0.23297,-0.22408]	0.001926 [0.001321,0.002552]
new density	0.23136 [0.22549,0.23709]	-0.23520 [-0.24241,-0.22808]	-0.003847 [-0.008807,0.000529]
legacy density	0.16834 [0.11436,0.20334]	-0.04459 [-0.19573,0.14779]	0.12375 [-0.00319,0.30034]

The masked new-density sum differs from the identically masked held-out sum by -0.005773 , or -2.306 bootstrap standard errors. The applicable density/trajectory gate therefore **FAIL**. The equilibrium masked comparison passes at 0.586 standard errors.

On the complete held-out driven trajectories, by contrast,

$$\langle J_{12} \rangle = 0.2327497, \quad \langle J_{23} \rangle = -0.2327327, \quad \langle J_{12} + J_{23} \rangle = 1.7046 \times 10^{-5},$$

with a whole-stream 95% interval $[-2.7013 \times 10^{-4}, 3.1681 \times 10^{-4}]$ and $z = 0.116$. The equilibrium complete-trajectory sum is 9.1214×10^{-5} with interval $[-3.1739 \times 10^{-4}, 5.3720 \times 10^{-4}]$ and $z = 0.411$. Thus the physical stationary current identity passes in both cases. We do not extrapolate the density outside its validated support to manufacture a global density result.

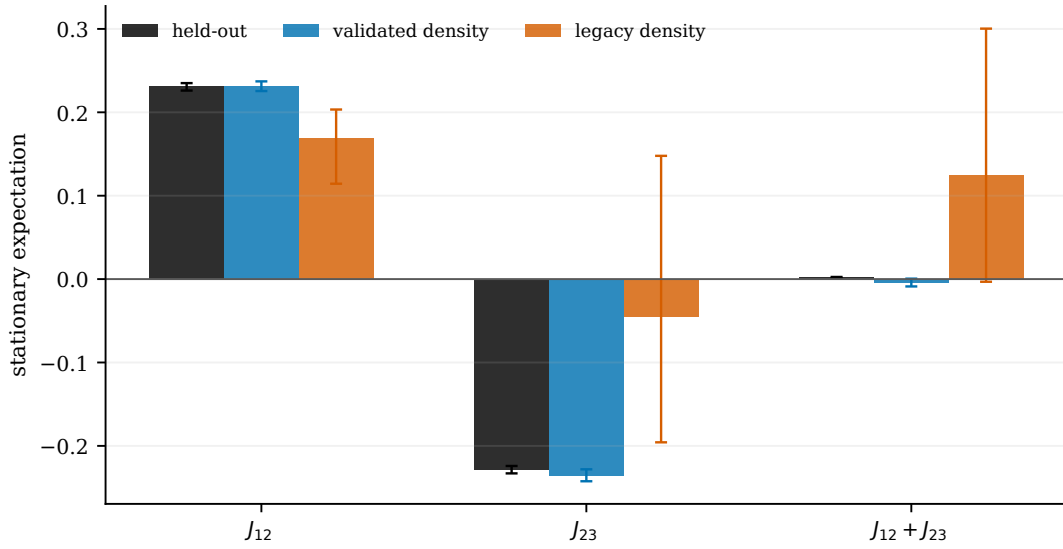


Figure 3: Current expectations on the Section 4.1 support mask. The new density reproduces the two directed currents far better than the legacy density, but its current sum remains 2.31 standard errors from the masked held-out value.

5 Gate ledger and claim boundary

Table 5: Frozen and scope-audited gates.

gate	observed	reference	result
minimum driven seed TV exceeds equilibrium upper floor	0.11405	0.04668	PASS
driven lower TV exceeds three times equilibrium upper floor	0.11369	0.14005	FAIL
phase density/trajectory interval overlaps	27/30	30/30	FAIL
masked current difference $ z \leq 1.96$	2.306	1.96	FAIL
global driven held-out current sum contains zero	1.70×10^{-5}	0	PASS

The legacy qualitative picture survives: the temperature gradient produces a reproducible endpoint-exchange asymmetry, phase locking strengthens with energy, and stationary transport balances the two bonds globally. The new density gives quantitatively closer estimates than the earlier unnormalized network, especially in the high-energy phase sector and for the directed currents. However, the rebuilt Section 4.2 is not fully density-validated under its frozen gates. Publication wording should distinguish direct trajectory results from density-derived reconstruction and explicitly report the three phase failures and the masked-current discrepancy. Claims apply only to the $\approx 99.4\%$ sampled-support region.

6 Reproducibility and provenance

No simulation or retraining was performed. The commands were

```
./run_analysis.sh
./run_current_scope_audit.sh
./run_report_summary.sh
```

from `experiments/short_chain_stabilization_density_rebuild_2026-09-15`. The source snapshot and analysis hashes are

```
source snapshot: 55c6670640bd7c686f1b3704dfa003c0b793ef4d
core analysis:   31c4288612631 added092dd6688f6267d9cf4fd9a5617a8ec57ac5f0269cdda192
scope audit:     d7e48ff24dcfec856e2743199bf24bd05c9a20b7eb4fc26d99b5a8405028c988
```